

Technical Roadmap: Real-Time Weapon Detection & Context-Aware Red Alert System

A Two-Person Collaborative Research Proposal

Project Domain: Computer Vision, Temporal Sequence Analysis, XAI

1 Project Abstract

Building upon the foundational work of previous academic terms (e.g., Driver Drowsiness and Dermal Classification), this project transitions from simple object classification to **Active Threat Intelligence**. By fusing the high-speed detection capabilities of the **YOLOv11/v12** architecture with the global contextual understanding of **Swin Transformers**, we aim to build a system that identifies firearms and bladed weapons while distinguishing them from benign handheld electronics. The project culminates in an automated **Red Alert System** that triggers visual, auditory, and digital notifications.

2 Phase 1: Dataset Synthesis & Pre-processing

To achieve high precision in cluttered environments, we will implement a multi-stage data pipeline:

- **Data Aggregation:** A composite dataset of $\approx 45,000$ images sourced from:
 - *University of Granada (SOHAS)*: Handgun and knife handling.
 - *Unidpro 2026*: High-definition staged tactical scenarios.
 - *Simuletic Synthetic*: Simulation of low-light and adverse weather CCTV feeds.
- **Hard Negative Mining:** Specifically labeling “Confuser Objects” (smartphones, wallets, drills, umbrellas) to force the model to learn fine-grained discriminative features.
- **Slicing Aided Hyper Inference (SAHI):** Implementation of a tiling strategy for inference. Given an image I , it is partitioned into overlapping patches $P_{i,j}$ of size $s \times s$:

$$I = \bigcup_{i,j} P_{i,j} \implies \text{Inference}(I) = \text{NMS}(\bigcup \text{Inference}(P_{i,j})) \quad (1)$$

This prevents the loss of feature resolution for small objects in 4K surveillance streams.

3 Phase 2: Hybrid Model Architecture

The system utilizes a state-of-the-art hybrid backbone to balance latency and accuracy.

- **Backbone:** YOLOv11 for local spatial feature extraction.
- **Neck:** **Swin Transformer** blocks using Shifted Windows (S-W-MSA) to capture long-range dependencies between the person and the potential weapon.
- **Loss Function:** We utilize a compound loss including CIoU for bounding box regression and **Focal Loss** for class imbalance:

$$L_{total} = \lambda_{box} L_{CIoU} + \lambda_{cls} [-\alpha_t (1 - p_t)^\gamma \log(p_t)] \quad (2)$$

4 Phase 3: Hand-Weapon Proximity & Threat Logic

An advanced alert should only fire if a weapon is being *actively handled*.

1. **Detection Heads:** The model will output two specific classes: **Hand** and **Weapon**.
2. **Proximity Condition:** Let B_h be the bounding box for a hand and B_w for a weapon. A threat T is validated if:

$$\text{Threat}(T) = \begin{cases} 1 & \text{if } \frac{\text{Area}(B_h \cap B_w)}{\text{Area}(B_h \cup B_w)} > \tau \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

where τ is the empirically determined Intersection over Union (IoU) threshold.

5 Phase 4: Multimodal Red Alert System

Upon threat validation, a prioritized alert protocol is executed:

- **Visual UI Alert:** A flashing red border on the GUI implemented via a frame-counter modulo:

$$\text{BorderColor} = \begin{cases} \text{Red} & \text{if } (\text{frame_idx} \bmod 10) < 5 \\ \text{Transparent} & \text{else} \end{cases} \quad (4)$$

- **Auditory Trigger:** Triggering a 1500Hz square wave alarm using the **Pygame** mixer.
- **Digital Notification:** Automated execution of an asynchronous **Telegram Bot API** request to send a JPEG snapshot and the GPS coordinates of the camera to security personnel.

6 Phase 5: Explainability & Optimization

- **XAI:** Using **Grad-CAM** to visualize the gradient flow. This ensures the model is activating on the *trigger guard* or *blade edge* rather than the carrier's clothing.
- **Quantization:** Conversion of the PyTorch weights to **TensorRT (INT8 precision)** to ensure the system maintains > 40 FPS on NVIDIA Jetson or RTX-series hardware.

7 Team Responsibility Matrix

Task Category	Partner A (Architect)	Partner B (Data/Alerts)
Model Development	Swin-Transformer Integration	Dataset Merging & Cleaning
Logic Layer	Proximity Detection Algorithms	SAHI Tiling Integration
Alert Systems	Telegram Bot & SMS Backend	OpenCV Visual/Audio Overlays
Validation	TensorRT Quantization	Grad-CAM & mAP Evaluation

8 Success Metrics

- **Primary:** $\text{mAP}@0.5:0.95 \geq 0.72$.
- **Secondary:** False Positive Rate (FPR) on "Confuser Objects" $< 2\%$.
- **Inference Speed:** $\leq 25\text{ms}$ per 1080p frame.